The transformer is an architecture that understands a sentence by having every word looked at directly at every other word, simultaneously and rebuilds itself from the ones that matter.

The key idea

Self attention, each word is built on 3 vectors:

Key - the label a word advertises about itself, what it matches against
Value - the actual content the word has
Query - what the current word is looking for right now

Attention is the selective focus of a word onto another word. The mechanism by which each word decides how much to concentrate on every other word, whilst building its own meaning.

Take the sentence:

the cat slept because it was tired.

And look at the word 'it'.

Dot product

We take the dot product of the query of the word doing the looking (it) with the key of the word being looked at (e.g. the).

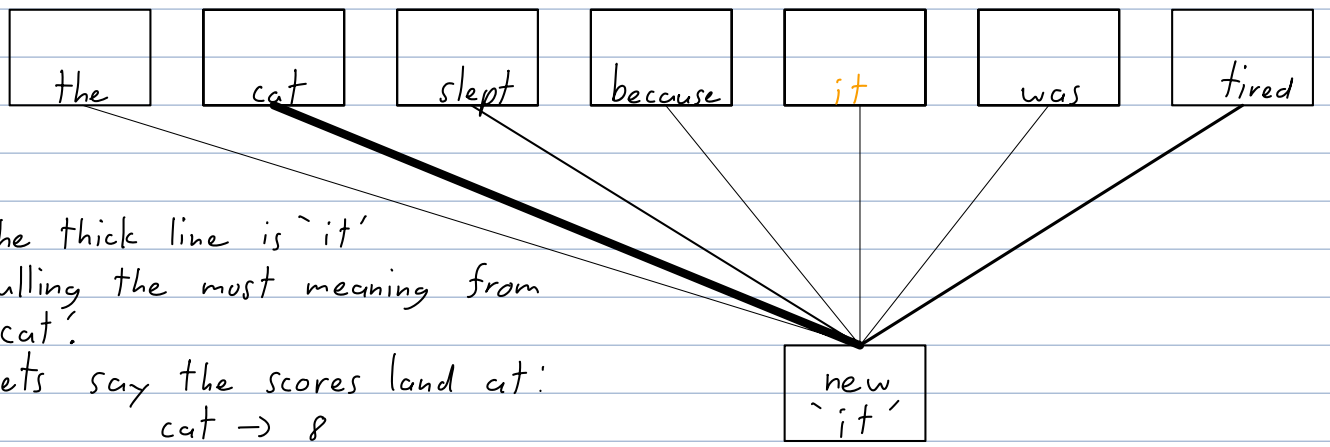
The dot product is used to output exactly one number for every pair of words, this works as all the information of the word is stored as a vector.

— query — key



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Weighting in action:



The thick line is `it' pulling the most meaning from `cat'.

Lets say the scores land at:

cat → 8
it → 2
tired → 5

Softmax

The raw scores go through a softmax which squashes all values to positive and all add up to 1. From [8, 2, 5] to [0.7, 0.05, 0.25]

Then the new value of `it' is = 0.7 V(cat) + 0.05 V(it) + 0.25 (tired)

